

Cognitive Abstraction Layers (CAL): A Research Architecture for Hierarchical Semantic Compression in AI Systems

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Abstract

Thesis: scalable cognition requires hierarchical semantic compression. More precisely: inference scalability depends on *semantic state topology*, not raw context size. The quality of AI governance scales with the structure of the compressed state space — with whether the compression preserves the topology of governance-relevant states — not with the number of tokens processed. As AI systems generate outputs at scales that exceed human supervisory capacity, the fundamental challenge of AI governance shifts from generation to comprehension. We propose that this challenge cannot be solved by increasing human review bandwidth or improving artifact-level tooling, because both responses operate at the wrong level of abstraction. We introduce the **Cognitive Abstraction Layer (CAL) architecture**: a five-level hierarchy spanning raw token streams (L0) to autonomous meta-inference on compressed semantic volumes (L4). The core invariant is *semantic conservation*: each layer must compress state while preserving sufficient causal and decisional structure for effective reasoning at the layer above. We formalize this requirement through **Semantic Information Density (SID)** — defined as the mutual information between the compressed representation and the governance decision space, normalized by the original. The central claim is that the quality of AI governance scales with SID_D at each layer boundary, not with context window size. L1 is instantiated by existing large language model inference but not formally structured as a governance layer; L2 is formalized and empirically validated by the companion paper TCO-L2 [1]; L3 and L4 represent open research directions with defined formal requirements and testable hypotheses.

Keywords: cognitive governance, semantic compression, tensor representation, hierarchical inference, mutual information, Natural Cognitive Frontier, Semantic Information Density, governance manifold

Contents

1	Introduction	3
1.1	Scope and Non-Goals	3

2	The CAL Architecture	4
3	L0–L1: Foundation Layers	4
3.1	L0: Token and Artifact Space	4
3.2	L1: Semantic Feature Layer	5
4	L2: Cognitive Tensor Layer — TCO-L2	5
4.1	Vectorization: φ — L1 Features $\rightarrow$ Quality Vector	5
4.2	Aggregation: f — Quality Vectors $\rightarrow$ Cognitive Tensor	5
4.3	Inference: I — Tensor $\rightarrow$ Governance Outputs	6
4.4	The Natural Cognitive Frontier (NCF)	6
4.5	L2 as the Empirical Anchor for Semantic Conservation	6
5	L3: Tensor Volume Layer	6
5.1	Formal Definition	7
5.2	Requirements for the Composition Operator C	7
5.3	Tractability and Mitigation Strategies	7
5.4	Open Research Questions at L3	8
5.5	Connections to Existing Research	8
5.6	The Governance Manifold Hypothesis	8
6	L4: Meta-Inference Layer	9
6.1	Formal Definition	9
6.2	The L4 Efficiency Hypothesis	9
6.3	Open Research Questions at L4	10
7	Semantic Information Density	10
7.1	Formal Definition	10
7.2	Operationalization at L2	11
7.3	The Open Problem: SID Without a Human Reference	11
8	Research Agenda	12
9	Positioning Within Existing Research	12
10	Conclusion	13

1 Introduction

The dominant paradigm in AI-assisted software development positions the human as a validator of AI-generated artifacts. As agent pipelines mature, this paradigm is reaching a structural limit: the volume of output generated by concurrent AI agents systematically exceeds the human capacity for artifact-level review. Analysis of code repositories with high proportions of AI-generated content documents increasing defect rates despite larger output volumes [2]. The bottleneck is not generation — it is the human capacity to maintain systemic comprehension over the outputs of multiple concurrent agents operating at different stages of a shared pipeline.

Two common responses to this problem are: (1) reduce the human’s role further, replacing review with automated quality checks, and (2) improve tooling around the existing paradigm — better diffs, richer IDEs, faster search. We argue that both responses are incomplete, because they operate at the wrong abstraction level. The problem is not that humans review too little, nor that their tools are inadequate. The problem is that the information presented to human supervisors is at the wrong abstraction level for the cognitive operations they are being asked to perform.

This paper proposes a third response: a hierarchical architecture of semantic abstraction layers that progressively compresses AI-generated state into forms appropriate for increasingly autonomous inference. We call this the **Cognitive Abstraction Layer (CAL) architecture**. It has five levels, L0 through L4. The human supervisor appears at Level 2 — not as a reviewer of artifacts, but as an orchestrator of system states. Below Level 2, generation and local inference are automated; above Level 2, governance inference is automated. At Level 2, the human operates at the **Natural Cognitive Frontier (NCF)** [1]: the abstraction level where cognitive demand is calibrated to human capacity, decisions are expressible in natural language, and judgment produces policy rather than correction.

The key question the architecture poses is not “how much can humans supervise?” but “what is the minimum semantic representation that preserves the structure required for effective governance at each level?” This question is formalized through **Semantic Information Density (SID)**: the quantity of decision-relevant cognitive structure preserved per unit of representation. The thesis of the CAL architecture is that maximizing SID_D at each layer boundary — not expanding context windows — is the path to scalable AI governance.

The current paper is a research agenda. We do not report new experimental results for L3 and L4. We provide: (a) formal definitions of each layer and the properties required at each boundary; (b) a summary of the L2 empirical anchor (TCO-L2); (c) concrete open problems with testable hypotheses for L3 and L4; and (d) the SID metric as a unifying measurement framework.

1.1 Scope and Non-Goals

CAL is a research architecture for AI pipeline governance in multi-agent software development contexts. It is *not*:

- **A theory of general cognition or AGI.** The word “cognitive” in CAL refers to the supervisory operations that agents and humans perform over pipeline state, not to cognition in the cognitive science or AGI sense.
- **A proposal to eliminate human governance.** L2 (TCO-L2) places the human orchestrator at the center of the architecture. L3 and L4 automate inference where human working memory

is the bottleneck — they do not eliminate human accountability.

- **A general inference efficiency claim.** The L4 Efficiency Hypothesis applies specifically to governance inference over semantically compressed pipeline state. It does not claim that hierarchical abstraction improves efficiency for arbitrary tasks where semantic compression may destroy rather than preserve relevant structure.
- **A complete, deployed system.** L3 and L4 have defined formal requirements and open problems, not implementations.

The scope is: multi-agent AI pipelines with measurable quality dimensions (as defined by φ in Section 4) and known pipeline structure. Extension to other domains requires domain-specific instantiation of φ and fresh validation of the L1→L2 interface.

2 The CAL Architecture

The CAL architecture is a five-level hierarchy. Each level receives compressed state from the level below, performs a defined transformation, and outputs a further-compressed representation to the level above. The architecture is bidirectional: state flows upward (L0→L4) and policy flows downward (L4→L0). At L2, the human orchestrator is the bidirectional interface.

Listing 1: CAL architecture overview

```

L4  Meta-Inference Layer      M(V) -> {decisions, predictions}
    | M: V -> structured governance outputs
L3  Tensor Volume Layer       V(T_1, T_2, ..., T_n)
    | C: {T} -> V      [composition operator -- open problem]
L2  Cognitive Tensor Layer    T[d,i,j,k]          <- TC0-L2
    | f o phi: {A} -> T,  I: T -> {Omega, Delta, Rho, Xi}
    | Human orchestration at NCF (natural language policy)
L1  Semantic Feature Layer    {embeddings, relations, signals}
    | LLM inference: tokens -> features [instantiated]
L0  Token / Artifact Space   raw AI-generated outputs

```

The core invariant at every boundary $L_k \rightarrow L_{k+1}$ is **semantic conservation**: the compressed representation at L_{k+1} must preserve sufficient causal and decisional structure for effective reasoning at L_{k+1} . A representation that fails this invariant destroys governance-relevant information that cannot be recovered at higher layers. Measuring the degree to which each boundary satisfies this invariant is the role of SID (Section 7).

3 L0–L1: Foundation Layers

3.1 L0: Token and Artifact Space

L0 is the space of raw AI-generated outputs: source code, configuration files, test results, documentation, deployment manifests, system logs. L0 is the entry point to the abstraction hierarchy and is not a contribution of CAL — it is the existing output of any AI-assisted development pipeline. Its key property is *volume asymmetry*: modern agentic pipelines generate artifacts at rates that exceed human review capacity [2], motivating the entire hierarchy.

3.2 L1: Semantic Feature Layer

L1 is the space of semantic features extracted from L0 artifacts by inference models: embeddings (dense vector representations of artifact semantics), relational features (dependency graphs, call graphs, architectural relationships), and quality signals (static analysis outputs, test coverage, linting results).

Critical clarification — L1 is instantiated, not formalized. Existing large language models generate L1-type features as intermediate representations in service of their primary generation and evaluation tasks. They do not expose a structured, persistent L1 state that formally serves as input to a governance layer. The CAL architecture does not propose building L1 from scratch — it formalizes the interface between L1 and L2, specifying what properties L1 outputs must have to support effective L2 governance.

Formally, L1 produces a set of per-artifact feature representations $\mathcal{F} = \{f_1, f_2, \dots, f_n\}$ where each f_i is a local representation of a single artifact. L1 does not perform cross-artifact, cross-temporal, or cross-agent aggregation — these are structurally the responsibility of L2.

4 L2: Cognitive Tensor Layer — TCO-L2

L2 is formalized and empirically validated by the companion paper TCO-L2 [1]. This section summarizes its contribution in the context of the CAL architecture.

4.1 Vectorization: φ — L1 Features $\rightarrow$ Quality Vector

$$\varphi : A \rightarrow \mathbf{V} \in [0, 1]^{11}$$

The vectorization function φ maps each artifact to an 11-dimensional quality vector $\mathbf{V} = (v_1, \dots, v_{11})$, spanning: functional correctness (v_1), architectural alignment (v_2), scalability projection (v_3), security risk (v_4 , inverted), observability coverage (v_5), testability (v_6), maintainability (v_7), technical debt (v_8 , inverted), performance (v_9), evaluator confidence (v_{10}), and anomaly score (v_{11} , inverted).

Epistemological note. Dimensions v_1, v_2, v_3, v_9 are *supervisory estimators* — heuristic semantic signals generated by LLM assessment without universal ground truth. Dimensions v_4, v_6, v_7, v_8 are *verifiable ground truth* — outputs of deterministic static analysis (Bandit, Radon). This distinction is critical for calibrating SID_D at the L1→L2 boundary.

4.2 Aggregation: f — Quality Vectors $\rightarrow$ Cognitive Tensor

$$f : \{\mathbf{V}\} \rightarrow T[d, i, j, k] \in \mathbb{R}^{n \times |S| \times |A| \times |T_{\text{idx}}|}$$

The aggregation function f builds a four-dimensional tensor indexed by quality dimension (d), pipeline stage (i), agent (j), and time index (k). The tensor’s shared index structure $[d, i, j, k]$ makes multidimensional supervisory operations first-class.

Two experimental scenarios demonstrate the operational necessity of this representation:

S3 — Gradual technical debt. Per-cycle degradation in v_8 of -0.08 falls within individual evaluator noise ($\sigma=0.07$, $\text{SNR}=1.14$). The 3-cycle cumulative delta (-0.24 , $\text{SNR}=3.43$) reliably crosses the alert threshold. Monte Carlo simulation ($n=1000$) confirms: artifact-level detection

rate 32.3% vs. tensor Δ detection rate 66.9% (+34.6 pp gap). The tensor operation $T[v_8, :, :, 0] - T[v_8, :, :, 3]$ is one line; the SQL equivalent requires a self-join reconstructing the coordinate system.

S5 — Inter-agent conflict. Two agents produce opposing quality profiles ($v_4 \Delta=0.65$, $v_6 \Delta=0.50$). Per-artifact review requires the supervisor to hold both outputs in working memory simultaneously. The tensor ϱ operation $|T[:, i, j_0, k] - T[:, i, j_1, k]|$ detects both conflicting dimensions automatically.

4.3 Inference: I — Tensor $\rightarrow$ Governance Outputs

$$I : T \rightarrow \{\Omega, \Delta, \varrho, \Xi\}$$

Symbol	Output	Threshold	Description
Ω	Global state	stable ≥ 0.70	stable / warning / critical
Δ	Trend	$ \text{slope} > 0.05$	Early-warning per dimension
ϱ	Systemic risk	$ \text{diff} > 0.30$	Inter-agent quality conflicts
Ξ	Recommendations	Top-3 Δ + top-2 ϱ	Ranked by estimated $\partial\Omega/\partial\text{action}$

4.4 The Natural Cognitive Frontier (NCF)

The NCF is the abstraction level at which cognitive demand is calibrated to human capacity: decisions do not exceed working memory bounds ($\leq 4\text{--}5$ chunks [3]), are expressible in natural language, and produce policy rather than artifact correction. At L2, the human orchestrator reads $\{\Omega, \Delta, \varrho, \Xi\}$ and injects policy P_{new} in natural language.

NCF operationalization in TCO-L2 uses four observable proxies: NASA Raw-TLX subscales (working memory saturation), correction log consistency σ across fault categories, accuracy variance $\sigma(\alpha)$ across tasks T1 $\rightarrow$ T4, and interaction timer IQR (attention fragmentation).

4.5 L2 as the Empirical Anchor for Semantic Conservation

The TCO-L2 experiment (between-subjects RCT, $n=40$ software engineers) tests whether participants make more accurate governance decisions from T than from raw L0 artifacts. This is not only an HCI claim — it is the empirical validation that the L0 $\rightarrow$ L2 abstraction is semantically conservative.

Dual purpose of the experiment. (1) Validate that the TCO-L2 dashboard reduces cognitive load and improves oversight quality. (2) Establish empirical evidence that tensor representation preserves sufficient decision-relevant structure for effective governance — the first measured data point for $\text{SID}_D(L0 \rightarrow L2)$. A positive result on H2 (Cohen’s $d > 0.50$) is equivalent to asserting $\text{SID}_D(L0 \rightarrow L2)$ above the baseline required for governance.

5 L3: Tensor Volume Layer

L3 is the first layer above human working memory in the CAL hierarchy. It operates on **tensor volumes $\mathbf{V}$** — collections of L2 cognitive tensors aggregated across sessions, pipelines, or time windows — without requiring the human as an inference substrate.

5.1 Formal Definition

Let $T^{(s)}$ denote the cognitive tensor for session s . A tensor volume is:

$$\mathbf{V} = C\left(\{T^{(s)}\}_{s=1}^n\right)$$

where C is the **composition operator** — the central open problem of L3. C maps a collection of session tensors into a unified higher-order structure preserving their causal and temporal relationships.

5.2 Requirements for the Composition Operator C

For $\mathbf{V}$ to be semantically conservative with respect to L4 inference, C must satisfy four properties:

- P1. Causal preservation.** Causal relationships between quality dimensions recorded within any $T^{(s)}$ must be encodeable in $\mathbf{V}$. Using Pearl’s do-calculus notation [4], the interventional distribution $P(d_j \mid \text{do}(d_i))$ must remain estimable from $\mathbf{V}$.
- P2. Temporal coherence.** The time indices of constituent tensors must be composable into a global temporal ordering. Cross-session drift must be representable in $\mathbf{V}$.
- P3. Dimensional stability.** The 11-dimensional quality vector structure must be preserved across the composition. L4 must be able to apply the same inference operations $(\Omega, \Delta, \varrho, \Xi)$ to slices of $\mathbf{V}$ that it applies to T .
- P4. Tractability.** C must be computable for continuous pipelines. The size of $\mathbf{V}$ must not grow linearly with n for large n .

5.3 Tractability and Mitigation Strategies

The composition operator C faces a combinatorial growth risk that must be treated as a primary design constraint. A naive tensor product $C = T^{(1)} \otimes T^{(2)} \otimes \dots \otimes T^{(n)}$ grows exponentially in dimensionality, violating Property P4 and negating the L4 Efficiency Hypothesis. Four mitigation strategies are candidates for the L3 research agenda:

1. **Sparse tensor representation.** If agent-stage-dimension combinations rarely co-occur, sparse formats (COO, CSR) reduce storage and computation to $\mathcal{O}(k)$ in the number of non-zero entries k , where $k \ll n^{11}$.
2. **Low-rank approximation.** Tucker and CP decompositions [5] approximate high-dimensional tensors as products of smaller factor matrices. If the effective rank of $\mathbf{V}$ is bounded by the number of distinct governance-relevant patterns, low-rank $\mathbf{V}$ is tractable.
3. **Selective retention (dynamic forgetting).** C can implement a selective update rule — analogous to gating in SSMs [6] — discarding low-SID_D sessions and retaining high-SID_D ones, keeping the effective n bounded.
4. **Manifold projection.** If governance-relevant states lie on a low-dimensional manifold (see Section 5.6), $\mathbf{V}$ can be projected onto this manifold with bounded information loss.

5.4 Open Research Questions at L3

- Does C belong to a known algebraic family (tensor product, contraction, CP/Tucker factorization)? What are the guarantees of each candidate?
- Does $\mathbf{V}$ exhibit stable *attractor states* representing semantic invariants of the pipeline?
- Does $\mathbf{V}$ exhibit *phase transitions* — discontinuous changes in global governance state analogous to regime shifts in complex systems?
- How does the effective rank of $\mathbf{V}$ grow with n ? Polynomial growth implies SID_D decreases with pipeline scale — a critical failure mode to characterize.

5.5 Connections to Existing Research

State Space Models. SSMs such as Mamba [6] demonstrate that temporal sequences can be compressed into fixed-size state representations with sub-quadratic complexity while preserving long-range dependencies via selective state spaces. The composition operator C in L3 is a semantic analog of the SSM state transition, operating on semantically structured quality tensors rather than raw token sequences.

Renormalization Group Theory. In statistical physics, renormalization reduces the degrees of freedom in a system while preserving macroscopic behavior [7]. L3 is a semantic coarse-graining: it reduces the dimensionality of the cognitive state space while preserving macroscopic governance properties (trends, conflicts, anomalies) that matter for L4 inference.

Tensor Decomposition. Tucker, CP, and tensor train/ring formats [5] offer tractable low-rank approximations and are candidates for C when exact composition is intractable.

Causal Representation Learning. The do-calculus requirement for C (Property P1) connects to causal representation learning [4]. Two methods are directly applicable: (a) *causal graph priors*, encoding known causal relationships between quality dimensions as structural constraints on C ; and (b) *causal regularization*, penalizing representations where interventional distributions $P(d_j \mid \text{do}(d_i))$ are not recoverable from $\mathbf{V}$. Causal graph priors derived from domain knowledge about software pipeline dynamics are the more tractable near-term approach, since causal discovery in compressed spaces is computationally expensive [4].

5.6 The Governance Manifold Hypothesis

The tractability mitigations in Section 5.3 are most effective when the following foundational hypothesis holds.

Hypothesis 1 (Governance Manifold). The set of governance-relevant pipeline states is a low-dimensional manifold $\mathcal{M}_{\text{gov}}$ embedded in the full tensor space $\mathbb{R}^{n \times |S| \times |A| \times |T_{\text{id}}|}$. The intrinsic dimensionality $\dim(\mathcal{M}_{\text{gov}})$ is bounded by the number of distinct governance-relevant failure patterns the pipeline can produce, not by the full tensor dimensionality.

If true, the consequences propagate through the entire architecture:

- **Tractability.** $\mathbf{V}$ needs only to represent $\mathcal{M}_{\text{gov}}$, not the ambient space. $\dim(\mathcal{M}_{\text{gov}})$ may be orders of magnitude smaller than the ambient dimension.

- **SID geometry.** $\text{SID}_D(L_k \rightarrow L_{k+1}) > \theta$ becomes a topological question: does C preserve the connectivity and local curvature of $\mathcal{M}_{\text{gov}}$?
- **L4 Efficiency.** If $M(\mathbf{V})$ operates on $\mathcal{M}_{\text{gov}}$, then $\text{Cost}(M(\mathbf{V})) = \mathcal{O}(\dim(\mathcal{M}_{\text{gov}}))$, decoupled from n . The L4 Efficiency Hypothesis becomes a consequence, not an independent claim.
- **Semantic collapse framing.** Collapse (Section 7.3) corresponds precisely to C reducing $\mathcal{M}_{\text{gov}}$ below its intrinsic dimensionality.

Testability at L2. The hypothesis is testable on the TCO-L2 corpus without new data collection. If quality vector trajectories across scenarios S1–S5 can be embedded in a low-dimensional space via UMAP [8] or Isomap while preserving inter-scenario distances, this supports a low-dimensional governance manifold at L2. The embedding dimensionality is the first measurable proxy for $\dim(\mathcal{M}_{\text{gov}})$.

This positions topology — not compression ratio — as the central mathematical object of CAL: the right question is not “by how much did we compress?” but “did we preserve the topology of the governance manifold?”

6 L4: Meta-Inference Layer

L4 performs inference on tensor volumes $\mathbf{V}$ rather than on individual tensors T or artifacts A . It is the fully automated inference layer: no human working memory is required as the inference substrate.

6.1 Formal Definition

$$M : \mathbf{V} \rightarrow \{\text{decisions, predictions, adaptations}\}$$

Because $\mathbf{V}$ (by Properties P1–P4 of C) encodes causal relationships, temporal coherence, and dimensional stability, M can perform reasoning not possible at lower layers:

- **Causal discovery:** identifying which quality dimensions causally influence others, across sessions.
- **Drift prediction:** predicting future quality trajectories from historical volume dynamics, before the signal crosses the L2 alert threshold.
- **Cross-pipeline conflict detection:** identifying conflicts between architectural strategies spanning multiple sessions.
- **Policy generation:** producing governance policies from historical patterns for human review at L2 or direct injection in high-confidence, low-stakes cases.

6.2 The L4 Efficiency Hypothesis

Hypothesis 2 (L4 Efficiency). There exists an inference architecture such that the cost of $M(\mathbf{V})$ scales with $\kappa(\mathbf{V})$ — the structural complexity of $\mathbf{V}$ (e.g., its effective rank, the entropy of its

attractor distribution, the size of its causal graph) — where $\kappa(\mathbf{V})$ grows significantly more slowly than $\mathcal{O}(n^2)$ in n (the raw artifact count of the underlying L0 space):

$$\text{Cost}(M(\mathbf{V})) = \mathcal{O}(\kappa(\mathbf{V})) \quad \text{where} \quad \kappa(\mathbf{V}) \ll \mathcal{O}(n^2).$$

This is not a claim that meta-inference is exponentially cheaper than token-level attention in all cases. It is a claim that if semantic abstraction is semantically conservative, inference cost can be decoupled from the volume of the original artifact stream and coupled instead to the structural complexity of the compressed governance state. This connects to the observation that current transformer architectures re-evaluate redundant relationships at every forward pass [9]; a hierarchy of semantic abstractions would eliminate this redundancy before inference.

Empirical testing requires: (a) a concrete definition of C and $\kappa(\mathbf{V})$; (b) a comparison of $M(\mathbf{V})$ cost vs. equivalent flat-context inference on raw artifacts; and (c) measurement of governance accuracy under both approaches. TCO-L2 provides the L2 baseline for (c).

6.3 Open Research Questions at L4

- What is the minimal $\mathbf{V}$ that supports M for a given class of governance decisions?
- Can M be trained end-to-end from labeled governance outcomes, or does it require explicit causal structure in $\mathbf{V}$ as supervision?
- What are the failure modes of M when C fails to preserve causal structure? How does governance accuracy degrade?
- At what scale (agents, sessions, dimensions) does $M(\mathbf{V})$ outperform equivalent flat-context LLM inference on governance tasks?

7 Semantic Information Density

The CAL architecture requires a metric applicable at every layer boundary to assess whether the abstraction is semantically conservative. Without such a metric, the claim that each layer “preserves decision-relevant structure” is unfalsifiable. We propose **Semantic Information Density (SID)**.

7.1 Formal Definition

Let R_k denote the representation at layer L_k , and let D be the set of governance decisions relevant to the pipeline. Define the **decision-relevant information content** of R_k as the mutual information between R_k and D :

$$I_{\text{decision}}(R_k) = I(D; R_k) = H(D) - H(D | R_k).$$

Definition 1 (Semantic Information Density).

$$\text{SID}_D(L_k \rightarrow L_{k+1}) = \frac{I(D; R_{k+1})}{I(D; R_k)} \in [0, 1].$$

A SID_D of 1.0 means R_{k+1} is as informative as R_k for governance decisions. A SID_D below threshold θ means the compression has destroyed governance-relevant information.

Mathematical guarantee: $SID_D \in [0, 1]$. By the data processing inequality [10], for the Markov chain $D \rightarrow R_k \rightarrow R_{k+1}$ (where R_{k+1} is produced by applying C to R_k):

$$I(D; R_{k+1}) \leq I(D; R_k),$$

guaranteeing $SID_D \leq 1$ for any valid compression, without additional assumptions. Information destroyed at a layer boundary cannot be recovered at higher layers — making SID_D a strict, irreversible measure of information loss.

Causal structure. A critical dimension of SID not captured by mutual information alone is causal structure preservation. A compression may preserve high SID_D for contemporaneous decisions while destroying the interventional distributions $P(d_j \mid \text{do}(d_i))$ needed for drift prediction and temporal governance — a silent failure mode.

SID is contextual, not absolute. Because $I(D; R_k)$ depends on the governance decision set D , SID is not a single scalar property — it is a family indexed by governance class:

$$SID_D(L_k \rightarrow L_{k+1}) \quad \text{for a specific governance class } D.$$

All SID claims in this paper are relative to the software development governance class defined by the TCO-L2 experimental scenarios (S1–S5). Extension to other domains requires fresh operationalization of D .

7.2 Operationalization at L2

The TCO-L2 experiment provides the first operational measurement of SID_D . Let α_2 be the governance accuracy of the experimental group (operating at L2) and α_0 the accuracy of the control group (operating at L0). A bounded estimate:

$$\widehat{SID}_D(L0 \rightarrow L2) \approx \frac{\alpha_2}{\alpha_{\max}},$$

where $\alpha_{\max}$ is the accuracy achievable with perfect information. H2 (Cohen’s $d > 0.50$) is equivalent to asserting $SID_D(L0 \rightarrow L2)$ substantially above baseline.

7.3 The Open Problem: SID Without a Human Reference

At L2, the human provides the calibration signal. At L3 and L4, this signal is unavailable. Three candidate approaches:

1. **Downstream outcome tracking.** Measure whether $M(\mathbf{V})$ decisions lead to measurable pipeline improvements over time.
2. **Synthetic benchmarks with known causal structure.** Generate pipelines with known ground-truth causal graphs; test whether $M(\mathbf{V})$ recovers the correct relationships.
3. **Comparison against L2 human decisions on held-out cases.** For novel governance scenarios, compare $M(\mathbf{V})$ against expert human decisions at L2 as a delayed reference.

Semantic collapse as a named failure mode. **Semantic collapse** occurs when C preserves acceptable SID_D for each individual governance class D while simultaneously destroying the diversity of the governance state space — collapsing distinctions between states relevant to *different* decision classes. This is analogous to mode collapse in generative models: local fidelity is maintained, global coverage is lost. Semantic collapse is particularly dangerous because it passes contemporaneous SID checks while producing systematic failures on novel or rare governance scenarios. Three mitigations: (a) *entropy preservation constraints*, requiring $H(\mathbf{V})$ to remain within a specified band of $H(T)$; (b) *attractor separation metrics*, requiring that distinct attractor states remain metrically separated in $\mathbf{V}$; (c) *governance diversity benchmarks*, test suites deliberately including rare failure modes to detect collapsed representations.

The Governance Manifold Hypothesis provides the geometric interpretation: collapse corresponds to C reducing $\dim(\mathcal{M}_{\text{gov}})$ below the intrinsic dimensionality of the governance manifold.

8 Research Agenda

Table 1: CAL research agenda by layer

Layer	Status	Target	Key open problem / testable hypothesis
L0	Production	—	Baseline; no open problem
L1	Instantiated	—	Formal L1→L2 interface; LLM signals have $SID_D > \theta$
L2	TCO-L2 (CHI 2027)	CHI 2027	NCF operationalization; H2: Cohen’s $d > 0.50$; $SID_D(L0 \rightarrow L2) > 0.70$
L3	Open	NeurIPS/ICML	Composition operator C (P1–P4); $SID_D(L2 \rightarrow L3) > \theta$
L4	Open	NeurIPS/ICML	M architecture; $\text{Cost}(M(\mathbf{V})) = \mathcal{O}(\kappa(\mathbf{V}))$ where $\kappa(\mathbf{V}) \ll \mathcal{O}(n^2)$
SID	Partial (L2)	Cross-layer	Measure SID_D without human reference; define D for non-software domains

9 Positioning Within Existing Research

AI Observability / MLOps. Datadog, New Relic, and OpenTelemetry monitor operational health (latency, errors, resources). TCO-L2 monitors semantic health (quality, debt, architectural alignment, inter-agent conflicts). CAL extends this to hierarchical governance.

Hierarchical Reinforcement Learning. HRL decomposes long-horizon tasks across temporal abstractions (options framework). CAL applies analogous hierarchical abstraction to governance, not to task execution.

State Space Models. SSMs [6] compress temporal sequences with sub-quadratic complexity. L3 requires a semantic analog of the SSM state transition for structured quality tensors.

Global Workspace Theory. GWT [11] proposes a shared workspace integrating specialized cognitive processes. The L2 tensor is a computational analog: a shared index structure integrating quality signals from multiple specialized evaluation agents.

Situation Awareness. Endsley’s SA model [12] (perception $\rightarrow$ comprehension $\rightarrow$ projection) maps to L0 $\rightarrow$ L1 $\rightarrow$ L2 in the CAL hierarchy. TCO-L2 is an operationalization of SA Levels 2–3 for AI pipeline supervision.

10 Conclusion

The CAL architecture proposes a reframing of the AI governance problem: not as a question of how much humans can supervise, but as a question of what abstraction level at each layer boundary maximizes SID_D for governance decisions. TCO-L2 is the empirical anchor — it validates, via controlled experiment, that the L0 $\rightarrow$ L2 abstraction is semantically conservative enough for effective human orchestration at the Natural Cognitive Frontier. That validation is the theoretical foundation for L3 and L4: if a human can govern effectively from a tensor representation, an automated inference layer operating on tensor volumes can do so without the human working memory constraint.

The central open problems — the composition operator C for L3, the measurement of SID_D without human reference, the L4 Efficiency Hypothesis, and the Governance Manifold Hypothesis — define a coherent research agenda with concrete testable hypotheses at each step. The claim is not that these problems are solved, but that they are the right problems to solve: progress on each one directly advances scalable AI governance.

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